

ME 505: Applied Finite Element Method

Course Code	ME 505
Course Name	Applied Finite Element Method
L-T-P-C	2-0-2-3
Prerequisites	ME 352 Finite Element Methods in Engineering
Intended for	UG/PG
Distribution	Elective
Approval	9th Senate (May 2016), in the M.Sc Applied Mathematics curriculum

Course Contents

• Module I

Introduction to FEM: Basics of statics, strength of materials and FEM, CAE driven design process, Analysis types: linear, non-linear, dynamic, buckling, thermal, Fatigue, optimization, CFD, NVH etc, 1D, 2D, 3D methods, Degree of freedom, Advantages of FEM, Modeling/Preprocessing techniques, introduction to meshing, common mistakes and errors, Application of analysis types in various engineering fields. [3 Lectures]

• FEM-Weighted Residue Approach

Non-weak type methods- methods adopted to minimize errors: Subdomain, Galerkin, Petrov-Galerkin, Least Square, Collocation; Weak form type method: Rayleigh-Ritz method, Finite element method, Global stiffness matrix, Shape functions, Direct application of element matrix equations, Compatibility, Convergence criteria, Sources of errors, Types of PDE: elliptic, parabolic and hyperbolic and their solution approach, 1D problems in heat transfer, fluid flow, vibration etc. and comparison with exact solution. [14 Lectures]

• Module II

Introduction to Meshing: 1-D Meshing- Introduction to meshing, when to use 1-D meshing, meshing in critical areas, element section, stiffness matrix derivation (direct method) and its properties, element types: beam element, rigid elements, fasteners, problems based on 1-D FEM and comparison with exact theory, 2-D Meshing: When to use 2-D elements, mid-surface, Constraint strain triangle, different types of element and their displacement function, Family of 2D elements: plane stress, plane strain, plate, membrane, thin shell etc., effect of mesh density, effect of biasing in critical region, boundary conditions, how not to mesh, shrink wrap meshing, problems based on 2-D FEM and comparison with exact theory. 3-D Meshing: When to use 3-D elements, DOF for solid elements, Algorithms, brick meshing, how not to mesh, Hexa and Penta elements, solid map meshing. [8 Lectures]

• Module III

Element Quality and Checks: Compatibility and mechanisms, spring elements, shells to solids, beam to solids, beams normal to shells, beam to shell edge, General element quality checks: skewness, aspect ratio, warpage, jacobian; 2-D quality checks, quality checks for tetra meshing, brick mesh quality checks, student projects on mesh quality, Weld, Bolt and Shrink Fit Modeling: Welding simulation-modelling spot and arc welding, bolted joints, bearing simulation, shrink fit simulation. [5 Lectures]

• Module V

Linear Static and Dynamic Analysis: Stiffness matrix, stress and strain calculations, FEM model for linear analysis, error analysis, design problems based on linear analysis, Theory of dynamic analysis: forced and free vibration, mode shapes, harmonic analysis, design for avoiding resonance, Thermal Analysis: Conduction, convection and radiation heat transfer, structured and unstructured meshing, IC engine block thermal analysis, Introduction to CFD, Nonlinear analysis: Introduction to nonlinearity, types of nonlinearity: geometric nonlinearity, material nonlinearity, boundary nonlinearity/contact nonlinearity, stress-strain measures, general procedures for nonlinear static analysis, plasticity.

Applied FEM: Projects based on thermal analysis, CFD, Fatigue analysis, NVH analysis, Crash analysis etc., application of FEA in biomedical, implant designs such as Orthopaedic Implants, Spine Implants, Cardiovascular Implants, medical device components, automotive, aerospace, civil etc. [12 Lectures]

Text Books

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References

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